# Type MS · Find named items in a program

## Answer 1 of 18 · Question 1 part d.iv · 1 mark

2025 JUN · Paper 02 · extract 1 of 5 · find part d.iv in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 1(a) | A1 | **The only correct answer is B**<br><br>***A*** *is not correct because for is used in iteration*<br>***C*** *is not correct because new is used to declare an object*<br>***D*** *is not correct because while is used in repetition* | | **1** |

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|
| 1(b)(i) | B1<br>B2<br>B3 | Award one mark for each correct cell:<br><br>**Description** / **Syntax error** / **Logic error** / **Runtime error**<br>The program does not produce the expected output. / / ✔ / <br>The program does not translate. / ✔ / / <br>The program crashes during execution. / / / ✔ | | **3** |

# Type MS · Find named items in a program

## Answer 1 of 18 · Question 1 part d.iv · 1 mark

2025 JUN · Paper 02 · extract 2 of 5 · find part d.iv in this mark scheme extract

Paper 2 – Application of Computational Thinking

| Question | mp | Answer | Additional guidance | Mark |
|---|---|---|---|---|
| 1(b)(ii) | B4 | Line 15 - : added to end of line | | |
| | B5 | Line 20 – 0 changed to 1 | | |
| | B6 | Line 22 – missing comma added between string and greenCount | | 3 |

```
# Q01bii

# initialise counters
ballCount = 0
greenCount = 0
redCount = 0

ballArray = ["green", "red", "green", "green", "red", "green","green", "green",
             "green", "green", "green", "red", "green", "red", "green", "red",
             "red", "green", "green", "red", "red", "green", "green", "red",
             "green", "red", "red", "green", "green", "red", "green", "green",
             "red", "green", "green", "green", "red", "green", "red","green"]

# iterate over ball array
for ball in ballArray:
    ballCount = ballCount + 1
    if ball == "green":
        greenCount = greenCount +1
    else:
        redCount = redCount + 1

print(" Green: ", greenCount, "balls \n", "Red: ", redCount,
      "balls \n", "Total: ", ballCount, "balls")
```

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# Type MS · Find named items in a program

## Answer 1 of 18 · Question 1 part d.iv · 1 mark

2025 JUN · Paper 02 · extract 3 of 5 · find part d.iv in this mark scheme extract

Paper 2 – Application of Computational Thinking

| Question | mp | Answer | Additional guidance | Mark |
|---|---|---|---|---|
| 1(c)(i) | C1 | Award one mark for:<br>• To store a value that must not be changed/prevent accidental change of value<br>• To simplify code maintenance/so that if the value must be updated the change only has to made in a single place<br>• To avoid inconsistency error of specifying same value multiple times<br>• To make code more readable/give meaningful names to values | Accidental change cannot refer to 'user' | 1 |

| Question | mp | Answer | Additional guidance | Mark |
|---|---|---|---|---|
| 1(c)(ii) | C2<br>C3 | Award **two** marks for a linked description such as:<br>• A variable stores a single value (1) whereas an array stores multiple values (1) | | 2 |

| Question | mp | Answer | Additional guidance | Mark |
|---|---|---|---|---|
| 1(d)(i) | D1 | Award **one** mark for:<br>• getWages (1) | Accept 'main'<br>Accept:<br>def getWages<br>def getWages (inSales, inAge)<br>line 7 (Python) | 1 |

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# Type MS · Find named items in a program

## Answer 1 of 18 · Question 1 part d.iv · 1 mark

2025 JUN · Paper 02 · extract 4 of 5 · find part d.iv in this mark scheme extract

Paper 2 – Application of Computational Thinking

| Question | mp | Answer | Additional guidance | Mark |
|---|---|---|---|---|
| 1(d)(ii) | D2 | Award **one** mark for:<br>• and (1) | Do not accept line numbers<br>Accept:<br>&& | 1 |

| Question | mp | Answer | Additional guidance | Mark |
|---|---|---|---|---|
| 1(d)(iii) | D3 | Award **one** mark for any of the following:<br>• sales (1)<br>• wages (1)<br>• inSales (1) | Do not accept line numbers | 1 |

| Question | mp | Answer | Additional guidance | Mark |
|---|---|---|---|---|
| 1(d)(iv) | D4 | Award **one** mark for any of the following:<br>• inSales (1)<br>• inAge (1) | Do not accept line numbers | 1 |

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## Type MS · Find named items in a program

### Answer 1 of 18 · Question 1 part d.iv · 1 mark

2025 JUN · Paper 02 · extract 5 of 5 · find part d.iv in this mark scheme extract

Paper 2 – Application of Computational Thinking

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

## Type MS · Find named items in a program

### Answer 2 of 18 · Question 2 part c.iii · 1 mark

2019 JUN · Paper 02 · extract 1 of 13 · find part c.iii in this mark scheme extract

| | | | |
|---|---|---|---|
| 2(b)(i) | 24.0 (1) | Accept 24 | (1) |

## Type MS · Find named items in a program

### Answer 2 of 18 · Question 2 part c.iii · 1 mark

2019 JUN · Paper 02 · extract 2 of 13 · find part c.iii in this mark scheme extract

| | | | |
|---|---|---|---|
| 2(b)(ii) | One of:<br>By the rules of precedence multiplication is done before addition. (1)<br>Without the brackets the addition would be done to the total of the multiplication. (1)<br>By the rules of the BODMAS/BIDMAS additions happen after multiplication. (1)<br>newSpeed and intialSpeed have to be added together before multiplying (by time). (1) | | (1) |

## Type MS · Find named items in a program

### Answer 2 of 18 · Question 2 part c.iii · 1 mark

2019 JUN · Paper 02 · extract 3 of 13 · find part c.iii in this mark scheme extract

| Question | Answer | Additional Guidance | Mark |
|---|---|---|---|

## Type MS · Find named items in a program

### Answer 2 of 18 · Question 2 part c.iii · 1 mark

2019 JUN · Paper 02 · extract 4 of 13 · find part c.iii in this mark scheme extract

| | | | |
|---|---|---|---|
| 2(b)(iii) | Real (1)<br>Because real x integer = real (1) | Accept double / float<br>Accept 'because it could be a number with decimal places' | (2) |

## Type MS · Find named items in a program

### Answer 2 of 18 · Question 2 part c.iii · 1 mark

2019 JUN · Paper 02 · extract 5 of 13 · find part c.iii in this mark scheme extract

| | | | |
|---|---|---|---|
| 2(b)(iv) | 27.375 (1) | | (1) |

## Type MS · Find named items in a program

### Answer 2 of 18 · Question 2 part c.iii · 1 mark

2019 JUN · Paper 02 · extract 6 of 13 · find part c.iii in this mark scheme extract

| | | | |
|---|---|---|---|
| Q2(c)(i) | C#: 16 or 20 (1)<br>Java: 16 or 21 (1)<br>Python: 8 or 10 (1) | | (1) |

## Type MS · Find named items in a program

### Answer 2 of 18 · Question 2 part c.iii · 1 mark

2019 JUN · Paper 02 · extract 7 of 13 · find part c.iii in this mark scheme extract

| | | | |
|---|---|---|---|
| Q2(c)(ii) | C#: 29 or 31 (1)<br>Java: 24 or 26 (1)<br>Python: 14 or 15 (1) | | (1) |

## Type MS · Find named items in a program

### Answer 2 of 18 · Question 2 part c.iii · 1 mark

2019 JUN · Paper 02 · extract 8 of 13 · find part c.iii in this mark scheme extract

| | | | |
|---|---|---|---|
| Q2(c)(iii) | Any of:<br>1. Message (1)<br>2. startPos (Python) / pos (Java or C#) (1)<br>3. characters (1) | | (1) |

## Type MS · Find named items in a program

### Answer 2 of 18 · Question 2 part c.iii · 1 mark

2019 JUN · Paper 02 · extract 9 of 13 · find part c.iii in this mark scheme extract

| | | | |
|---|---|---|---|
| Q2(c)(iv) | newMessage (1) | Do not accept outMessage | (1) |

## Type MS · Find named items in a program

### Answer 2 of 18 · Question 2 part c.iii · 1 mark

2019 JUN · Paper 02 · extract 10 of 13 · find part c.iii in this mark scheme extract

| | | | |
|---|---|---|---|
| Q2(c)(v) | Response starts with 'c' (1)<br>Response has six letters in sequence from 'educational' (1) | Marks can be awarded from Q2(c)(vi) | (2) |

## Type MS · Find named items in a program

### Answer 2 of 18 · Question 2 part c.iii · 1 mark

2019 JUN · Paper 02 · extract 11 of 13 · find part c.iii in this mark scheme extract

| | | | |
|---|---|---|---|
| Q2(c)(vi) | 1. Returns substring / crops a string (1)<br>2. Located between the two input values / with number of characters identified in the last parameter (characters) and starting at the second parameter (startPos) (1) | Award two marks for:<br>It produces a substring of the message starting at position startPos with length characters<br>Award up to 2 marks for a correct interpretation of the code (e.g 'obtain a newMessage' – is worth 1)<br>Accept a correct generic explanation of what a subprogram does for 1 mark | (2) |

## Type MS · Find named items in a program

### Answer 2 of 18 · Question 2 part c.iii · 1 mark

2019 JUN · Paper 02 · extract 12 of 13 · find part c.iii in this mark scheme extract

| Question | Answer | Additional Guidance | Mark |
|---|---|---|---|

## Type MS · Find named items in a program

### Answer 2 of 18 · Question 2 part c.iii · 1 mark

2019 JUN · Paper 02 · extract 13 of 13 · find part c.iii in this mark scheme extract

| | | | |
|---|---|---|---|
| 2(a) | Award 1 mark for each of:<br>1. set counter to 1 and guess to 0 (1)<br>2. set answer to random value between 1 and 10 (1)<br>3. request input of and accept an integer value for guess (1)<br>4. create a while loop with a correct condition (1)<br>5. increment the counter (1)<br>6. use if else selection to determine and display appropriate output message for incorrect guesses (1)<br>7. request input of and accept another integer value for a guess inside the loop (1)<br>8. display correct guess and count of guesses (1)<br>9. compiling without syntax errors (1)<br>10. coding meets all requirements of question (1) | Candidates are required to open the file Q02a in the code editor.<br>Amended code should be saved as Q02aFINISHED.<br>Logic of algorithm must be followed as set out.<br>Alternatives must address each point.<br>Do not penalise candidates who attempt more than the stated requirements. | (10) |

| Question | Answer | Additional Guidance | Mark |
|---|---|---|---|

## Type MS · Find named items in a program

### Answer 3 of 18 · Question 2 part c.iv · 1 mark

2019 JUN · Paper 02 · extract 1 of 13 · find part c.iv in this mark scheme extract

| | | | |
|---|---|---|---|
| 2(b)(i) | 24.0 (1) | Accept 24 | (1) |

## Type MS · Find named items in a program

### Answer 3 of 18 · Question 2 part c.iv · 1 mark

2019 JUN · Paper 02 · extract 2 of 13 · find part c.iv in this mark scheme extract

| | | | |
|---|---|---|---|
| 2(b)(ii) | One of:<br>By the rules of precedence multiplication is done before addition. (1)<br>Without the brackets the addition would be done to the total of the multiplication. (1)<br>By the rules of the BODMAS/BIDMAS additions happen after multiplication. (1)<br>newSpeed and intialSpeed have to be added together before multiplying (by time). (1) | | (1) |

## Type MS · Find named items in a program

### Answer 3 of 18 · Question 2 part c.iv · 1 mark

2019 JUN · Paper 02 · extract 3 of 13 · find part c.iv in this mark scheme extract

| Question | Answer | Additional Guidance | Mark |
|---|---|---|---|

## Type MS · Find named items in a program

### Answer 3 of 18 · Question 2 part c.iv · 1 mark

2019 JUN · Paper 02 · extract 4 of 13 · find part c.iv in this mark scheme extract

| | | | |
|---|---|---|---|
| 2(b)(iii) | Real (1)<br>Because real x integer = real (1) | Accept double / float<br>Accept 'because it could be a number with decimal places' | (2) |

## Type MS · Find named items in a program

### Answer 3 of 18 · Question 2 part c.iv · 1 mark

2019 JUN · Paper 02 · extract 5 of 13 · find part c.iv in this mark scheme extract

| | | | |
|---|---|---|---|
| 2(b)(iv) | 27.375 (1) | | (1) |

## Type MS · Find named items in a program

### Answer 3 of 18 · Question 2 part c.iv · 1 mark

2019 JUN · Paper 02 · extract 6 of 13 · find part c.iv in this mark scheme extract

| | | | |
|---|---|---|---|
| Q2(c)(i) | C#: 16 or 20 (1)<br>Java: 16 or 21 (1)<br>Python: 8 or 10 (1) | | (1) |

## Type MS · Find named items in a program

### Answer 3 of 18 · Question 2 part c.iv · 1 mark

2019 JUN · Paper 02 · extract 7 of 13 · find part c.iv in this mark scheme extract

| | | | |
|---|---|---|---|
| Q2(c)(ii) | C#: 29 or 31 (1)<br>Java: 24 or 26 (1)<br>Python: 14 or 15 (1) | | (1) |

## Type MS · Find named items in a program

### Answer 3 of 18 · Question 2 part c.iv · 1 mark

2019 JUN · Paper 02 · extract 8 of 13 · find part c.iv in this mark scheme extract

| | | | |
|---|---|---|---|
| Q2(c)(iii) | Any of:<br>1. Message (1)<br>2. startPos (Python) / pos (Java or C#) (1)<br>3. characters (1) | | (1) |

## Type MS · Find named items in a program

### Answer 3 of 18 · Question 2 part c.iv · 1 mark

2019 JUN · Paper 02 · extract 9 of 13 · find part c.iv in this mark scheme extract

| | | | |
|---|---|---|---|
| Q2(c)(iv) | newMessage (1) | Do not accept outMessage | (1) |

## Type MS · Find named items in a program

### Answer 3 of 18 · Question 2 part c.iv · 1 mark

2019 JUN · Paper 02 · extract 10 of 13 · find part c.iv in this mark scheme extract

| | | | |
|---|---|---|---|
| Q2(c)(v) | Response starts with 'c' (1)<br>Response has six letters in sequence from 'educational' (1) | Marks can be awarded from Q2(c)(vi) | (2) |

## Type MS · Find named items in a program

### Answer 3 of 18 · Question 2 part c.iv · 1 mark

2019 JUN · Paper 02 · extract 11 of 13 · find part c.iv in this mark scheme extract

| | | | |
|---|---|---|---|
| Q2(c)(vi) | 1. Returns substring / crops a string (1)<br>2. Located between the two input values / with number of characters identified in the last parameter (characters) and starting at the second parameter (startPos) (1) | Award two marks for:<br>It produces a substring of the message starting at position startPos with length characters<br>Award up to 2 marks for a correct interpretation of the code (e.g 'obtain a newMessage' – is worth 1)<br>Accept a correct generic explanation of what a subprogram does for 1 mark | (2) |

## Type MS · Find named items in a program

### Answer 3 of 18 · Question 2 part c.iv · 1 mark

2019 JUN · Paper 02 · extract 12 of 13 · find part c.iv in this mark scheme extract

| Question | Answer | Additional Guidance | Mark |
|---|---|---|---|

## Type MS · Find named items in a program

### Answer 3 of 18 · Question 2 part c.iv · 1 mark

2019 JUN · Paper 02 · extract 13 of 13 · find part c.iv in this mark scheme extract

| | | | |
|---|---|---|---|
| 2(a) | Award 1 mark for each of:<br>1. set counter to 1 and guess to 0 (1)<br>2. set answer to random value between 1 and 10 (1)<br>3. request input of and accept an integer value for guess (1)<br>4. create a while loop with a correct condition (1)<br>5. increment the counter (1)<br>6. use if else selection to determine and display appropriate output message for incorrect guesses (1)<br>7. request input of and accept another integer value for a guess inside the loop (1)<br>8. display correct guess and count of guesses (1)<br>9. compiling without syntax errors (1)<br>10. coding meets all requirements of question (1) | Candidates are required to open the file Q02a in the code editor.<br>Amended code should be saved as Q02aFINISHED.<br>Logic of algorithm must be followed as set out.<br>Alternatives must address each point.<br>Do not penalise candidates who attempt more than the stated requirements. | (10) |

| Question | Answer | Additional Guidance | Mark |
|---|---|---|---|

## Type MS · Find named items in a program

### Answer 4 of 18 · Question 1 part e.iii · 1 mark

2020 NOV · Paper 02 · extract 1 of 10 · find part e.iii in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 1 (a) | A1 | **C** (equals) (1) | | (1) |

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

Type MS · Find named items in a program

Answer 4 of 18 · Question 1 part e.iii · 1 mark

2020 NOV · Paper 02 · extract 2 of 10 · find part e.iii in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 1 (b) | B1 | Award one mark for any of:<br><br>• TRUE **and** FALSE (1)<br>• Yes **and** No (1)<br>• T **and** F (1)<br>• 0 **and** 1 (1) | Ignore capitalisation | (1) |

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

Type MS · Find named items in a program

Answer 4 of 18 · Question 1 part e.iii · 1 mark

2020 NOV · Paper 02 · extract 3 of 10 · find part e.iii in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 1 (d) | D1 | Award one mark for any of:<br>• (Symbolic name associated with) a value that may be changed (1)<br>• A container used to store data (1)<br>• A data store whose contents can change whilst a program is executing (1) | | (1) |

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

## Type MS · Find named items in a program

### Answer 4 of 18 · Question 1 part e.iii · 1 mark

2020 NOV · Paper 02 · extract 4 of 10 · find part e.iii in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 1 (e) (i) | E1 | Python 6<br>C# 12<br>Java 8 | | (1) |

## Type MS · Find named items in a program

### Answer 4 of 18 · Question 1 part e.iii · 1 mark

2020 NOV · Paper 02 · extract 5 of 10 · find part e.iii in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 1 (e) (ii) | E2 | Python 3<br>C# 7<br>Java 3 | | (1) |

Type MS · Find named items in a program

Answer 4 of 18 · Question 1 part e.iii · 1 mark

2020 NOV · Paper 02 · extract 6 of 10 · find part e.iii in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 1 (e) (iii) | E3 | Award one mark for any of:<br>• amount (1)<br>• x (1)<br>• total (1)<br>• count (1) | Accept listNumbers or numArray | (1) |

## Type MS · Find named items in a program

### Answer 4 of 18 · Question 1 part e.iii · 1 mark

2020 NOV · Paper 02 · extract 7 of 10 · find part e.iii in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 1 (e) (iv) | E4 | message | | (1) |

## Type MS · Find named items in a program

### Answer 4 of 18 · Question 1 part e.iii · 1 mark

2020 NOV · Paper 02 · extract 8 of 10 · find part e.iii in this mark scheme extract

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

## Type MS · Find named items in a program

### Answer 4 of 18 · Question 1 part e.iii · 1 mark

2020 NOV · Paper 02 · extract 9 of 10 · find part e.iii in this mark scheme extract

| | | | | | |
|---|---|---|---|---|---|
| 1 (c) | C1 | 2.1.5 | Delete space in variable name (1) | | (3) |
| | C2 | 2.1.5 | Capitalise O in variable name (1) | | |
| | C3 | 2.1.5 | Correction to print statement (1) | | |
| Code example | | | | | |
| Python | | | | | |

```
numOne = 25
numTwo = 36
numThree = numOne * numTwo
print (numThree)
```

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## Type MS · Find named items in a program

### Answer 4 of 18 · Question 1 part e.iii · 1 mark

2020 NOV · Paper 02 · extract 10 of 10 · find part e.iii in this mark scheme extract

| Question | mp | ref | Answer | Additional Guidance | Mark |
|---|---|---|---|---|---|

## Type MS · Find named items in a program

### Answer 5 of 18 · Question 1 part e.iv · 1 mark

2020 NOV · Paper 02 · extract 1 of 10 · find part e.iv in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 1 (a) | A1 | **C** (equals) (1) | | (1) |

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

Type MS · Find named items in a program

Answer 5 of 18 · Question 1 part e.iv · 1 mark

2020 NOV · Paper 02 · extract 2 of 10 · find part e.iv in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 1 (b) | B1 | Award one mark for any of:<br><br>• TRUE **and** FALSE (1)<br>• Yes **and** No (1)<br>• T **and** F (1)<br>• 0 **and** 1 (1) | Ignore capitalisation | (1) |

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

Type MS · Find named items in a program

Answer 5 of 18 · Question 1 part e.iv · 1 mark

2020 NOV · Paper 02 · extract 3 of 10 · find part e.iv in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 1 (d) | D1 | Award one mark for any of:<br>• (Symbolic name associated with) a value that may be changed (1)<br>• A container used to store data (1)<br>• A data store whose contents can change whilst a program is executing (1) | | (1) |

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

## Type MS · Find named items in a program

### Answer 5 of 18 · Question 1 part e.iv · 1 mark

2020 NOV · Paper 02 · extract 4 of 10 · find part e.iv in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 1 (e) (i) | E1 | Python 6<br>C# 12<br>Java 8 | | (1) |

## Type MS · Find named items in a program

### Answer 5 of 18 · Question 1 part e.iv · 1 mark

2020 NOV · Paper 02 · extract 5 of 10 · find part e.iv in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 1 (e) (ii) | E2 | Python 3<br>C# 7<br>Java 3 | | (1) |

Type MS · Find named items in a program

Answer 5 of 18 · Question 1 part e.iv · 1 mark

2020 NOV · Paper 02 · extract 6 of 10 · find part e.iv in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 1 (e) (iii) | E3 | Award one mark for any of:<br>• amount (1)<br>• x (1)<br>• total (1)<br>• count (1) | Accept listNumbers or numArray | (1) |

## Type MS · Find named items in a program

### Answer 5 of 18 · Question 1 part e.iv · 1 mark

2020 NOV · Paper 02 · extract 7 of 10 · find part e.iv in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 1 (e) (iv) | E4 | message | | (1) |

4 of 16

## Type MS · Find named items in a program

### Answer 5 of 18 · Question 1 part e.iv · 1 mark

2020 NOV · Paper 02 · extract 8 of 10 · find part e.iv in this mark scheme extract

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

## Type MS · Find named items in a program

### Answer 5 of 18 · Question 1 part e.iv · 1 mark

2020 NOV · Paper 02 · extract 9 of 10 · find part e.iv in this mark scheme extract

| | | | | | |
|---|---|---|---|---|---|
| 1 (c) | C1 | 2.1.5 | Delete space in variable name (1) | | (3) |
| | C2 | 2.1.5 | Capitalise O in variable name (1) | | |
| | C3 | 2.1.5 | Correction to print statement (1) | | |
| Code example | | | | | |
| Python | | | | | |

```
numOne = 25
numTwo = 36
numThree = numOne * numTwo
print (numThree)
```

9 of 16

## Type MS · Find named items in a program

### Answer 5 of 18 · Question 1 part e.iv · 1 mark

2020 NOV · Paper 02 · extract 10 of 10 · find part e.iv in this mark scheme extract

| Question | mp | ref | Answer | Additional Guidance | Mark |
|---|---|---|---|---|---|

## Type MS · Find named items in a program

### Answer 6 of 18 · Question 3 part d.ii · 1 mark

2021 JUN · Paper 02 · extract 1 of 8 · find part d.ii in this mark scheme extract

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|
| 3 (b) | B1 | Award up to 2 marks for a linked explanation such as:<br>• The number of keys are limited/only one shift used (1) making it easy to use brute force to decrypt (1)<br>• It can be easy to find commonly used letters (e.g. E) (1) and guess the key (1) | Accept alternative similar wording. | (2) |

## Type MS · Find named items in a program

### Answer 6 of 18 · Question 3 part d.ii · 1 mark

2021 JUN · Paper 02 · extract 2 of 8 · find part d.ii in this mark scheme extract

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|
| 3 (c) | | Award 1 mark each up to a maximum of 4 for:<br><table><tr><td>Encrypted letter</td><td>f</td><td>l</td><td>m</td><td>k</td><td>t</td><td>r</td><td>w</td><td>h</td><td>e</td><td>e</td></tr><tr><td>Keyword letter</td><td>t</td><td>h</td><td>i</td><td>r</td><td>t</td><td>y</td><td>t</td><td>h</td><td>i</td><td>r</td></tr><tr><td>Decrypted letter</td><td>m</td><td>e</td><td>e</td><td>t</td><td>a</td><td>t</td><td>d</td><td>a</td><td>w</td><td>n</td></tr></table> | | |
| | C1 | Award 1 mark from:<br>• flmktrwhee mapped to thirtythir (1)<br>• Flmktr mapped to thirty (1) | | |
| | C2 | At least one letter decrypted correctly (1) | | |
| | C3 | At least one word decrypted correctly (1) | | |
| | C4 | Decrypted message 'meet at dawn' (1) | | (4) |

## Type MS · Find named items in a program

### Answer 6 of 18 · Question 3 part d.ii · 1 mark

2021 JUN · Paper 02 · extract 3 of 8 · find part d.ii in this mark scheme extract

| Question | IMP | Answer | Additional Guidance | Mark |
|---|---|---|---|---|
| 3 (d)(i) | D1 | Award 1 mark for:<br>• cipher letter / a single encrypted letter (1) | Do not accept word/message/text | (1) |

## Type MS · Find named items in a program

### Answer 6 of 18 · Question 3 part d.ii · 1 mark

2021 JUN · Paper 02 · extract 4 of 8 · find part d.ii in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| | | • cipherLetter / a single encrypted letter (1) | | (1) |
| 3 (d)(ii) | D2 | Award 1 mark for any of:<br>• keywordLetter<br>• plaintextLetter | Ignore case | (1) |

## Type MS · Find named items in a program

### Answer 6 of 18 · Question 3 part d.ii · 1 mark

2021 JUN · Paper 02 · extract 5 of 8 · find part d.ii in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| | | • plaintextletter | | (1) |
| 3 (d)(iii) | D2 | Award 1 mark for any of:<br>• subprogram that is already defined<br>• subprogram that is already written<br>• subprogram that is already compiled<br>• subprogram that can be called without having to write code for it | Must be clear they are referring to subprogram provided by the language itself | (1) |

## Type MS · Find named items in a program

### Answer 6 of 18 · Question 3 part d.ii · 1 mark

2021 JUN · Paper 02 · extract 6 of 8 · find part d.ii in this mark scheme extract

## Type MS · Find named items in a program

### Answer 6 of 18 · Question 3 part d.ii · 1 mark

2021 JUN · Paper 02 · extract 7 of 8 · find part d.ii in this mark scheme extract

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|
| 3 (a) | A1 | Get plaintext and store in plaintext variable | Accept alternative wording<br>Line numbers may be different compared to the examples shown<br><br>When testing the completed code use lowercase for the input | (1) |
| | A2 | Get key and store in key variable | | (1) |
| | A3 | Validate key (must repeat until a valid key has been input) | | (1) |
| | A4 | Open file to write/append | | (1) |
| | A5 | Write cipher text | | (1) |
| | A6 | Close file | | (1) |
| | A7 | Displays ciphertext | | (1) |
| | A8 | Executing and producing correct output to file and screen (must have A3) | | (1) |
| Code examples | | | | |
| Python | | | | |

```
#       Add your code to get the plaintext and convert it to lowercase
plaintext = input("Enter the plaintext using lowercase letters: ")
plaintext = plaintext.lower()
```

```
# Add your code to get the key and make sure the key is between 1 and 25
while key < 1 or key > 25:
    key = int(input("Enter the key - a number between 1 and 25 "))
```

```
# Add your code to write the ciphertext to a text file
cipherFile = open("Cipher.txt", "w")
cipherFile.write(ciphertext)
cipherFile.close()
```

```
#       Add your code to display the ciphertext
print("The ciphertext message is:",ciphertext)
```

## Type MS · Find named items in a program

### Answer 6 of 18 · Question 3 part d.ii · 1 mark

2021 JUN · Paper 02 · extract 8 of 8 · find part d.ii in this mark scheme extract

## Type MS · Find named items in a program

### Answer 7 of 18 · Question 3 part a.i · 1 mark

2021 NOV · Paper 02 · extract 1 of 11 · find part a.i in this mark scheme extract

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|
| 3 (a) (i) | A1 | C# and Python monthNumber/monthsAndDays/count/found (1)<br>Java monthNumber/monthsAndDays/count/found/input | | (1) |

## Type MS · Find named items in a program

### Answer 7 of 18 · Question 3 part a.i · 1 mark

2021 NOV · Paper 02 · extract 2 of 11 · find part a.i in this mark scheme extract

| | | | |
|---|---|---|---|
| 3 (a) (ii) | A2 | monthsAndDays (1) | (1) |

## Type MS · Find named items in a program

### Answer 7 of 18 · Question 3 part a.i · 1 mark

2021 NOV · Paper 02 · extract 3 of 11 · find part a.i in this mark scheme extract

| | | | |
|---|---|---|---|
| 3 (a) (iii) | A3 | C#<br>• WriteLine, ToString, ToInt32, ReadLine<br>Java<br>• Print, ParseInt, nextLine, ToString, Equals<br>Python<br>• print/len/int/input (1) | (1) |

## Type MS · Find named items in a program

### Answer 7 of 18 · Question 3 part a.i · 1 mark

2021 NOV · Paper 02 · extract 4 of 11 · find part a.i in this mark scheme extract

| | | | |
|---|---|---|---|
| 3 (a) (iv) | A4 | Display/showMonthNameAndDays (1) | (1) |

## Type MS · Find named items in a program

### Answer 7 of 18 · Question 3 part a.i · 1 mark

2021 NOV · Paper 02 · extract 5 of 11 · find part a.i in this mark scheme extract

| | | | |
|---|---|---|---|
| 3 (a) (v) | A5 | pMonth (1) | (1) |

## Type MS · Find named items in a program

### Answer 7 of 18 · Question 3 part a.i · 1 mark

2021 NOV · Paper 02 · extract 6 of 11 · find part a.i in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 3 (a) (vi) | A6<br>A7 | **Award 1 mark for each of:**<br><br>**Type of test data** / **Test data**<br>Boundary (1) / 1<br>Erroneous (1) / 15 | | (2) |

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

## Type MS · Find named items in a program

### Answer 7 of 18 · Question 3 part a.i · 1 mark

2021 NOV · Paper 02 · extract 7 of 11 · find part a.i in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 3 (b) | B1<br>B2 | **Award up to 2 marks for a linked explanation such as:**<br><br>• A library program would not know the names of the calling program's variables in advance (1) meaning it would not be a reusable solution (1)<br>• (Using parameters) enables reusable solutions (1) using different data values/arguments (1)<br>• Using parameters uses less memory / uses memory more efficiently (1) because it avoids the use of global variables / because the memory is freed after the subprogram is finished (1) | | (2) |

## Type MS · Find named items in a program

### Answer 7 of 18 · Question 3 part a.i · 1 mark

2021 NOV · Paper 02 · extract 8 of 11 · find part a.i in this mark scheme extract

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

## Type MS · Find named items in a program

### Answer 7 of 18 · Question 3 part a.i · 1 mark

2021 NOV · Paper 02 · extract 9 of 11 · find part a.i in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 3 (c) | **Award 1 mark for each of:** | | | |
| | C1 | Condition to check whether the year exactly divides by 4 or equivalent condition (1) | | |
| | C2 | Condition to check whether year exactly divides by 400 or equivalent condition (1) | | |
| | C3 | Condition to check whether year does not exactly divide by 100 or equivalent condition (1) | | |
| | C4 | Correct message for at least one of the conditions (1) | | |
| | C5 | Program executes and produces the correct output (1) | | |
| | C6 | Efficient solution (1) | | |
| | | | | (6) |
| Code examples | | | | |
| C# | ```<br>if (year % 400 == 0)<br>{<br>Console.WriteLine(year + " is a leap year");<br>}<br>else if (year % 4 == 0 && year % 100 !=0)<br>{<br>Console.WriteLine(year + " is a leap year");<br>}<br>else<br>{<br>Console.WriteLine(year + " is not a leap year");<br>}<br>// End of main program<br>``` | | | |
| Java | ```<br>if (year % 400 == 0)<br>{<br>System.out.print(year + " is a leap year");<br>}<br>else if (year % 4 == 0 && year % 100 !=0)<br>{<br>System.out.print(year + " is a leap year");<br>}<br>else<br>{<br>System.out.print(year + " is not a leap year");<br>}<br>``` | | | |

## Type MS · Find named items in a program

### Answer 7 of 18 · Question 3 part a.i · 1 mark

2021 NOV · Paper 02 · extract 10 of 11 · find part a.i in this mark scheme extract

| | |
|---|---|
| Python | `if year % 400 == 0:`<br>`  print(year,"is a leap year")`<br>`elif year % 4 == 0 and year % 100 !=0:`<br>`  print(year,"is a lear year")`<br>`else:`<br>`  print(year,"is not a leap year")` |

## Type MS · Find named items in a program

### Answer 7 of 18 · Question 3 part a.i · 1 mark

2021 NOV · Paper 02 · extract 11 of 11 · find part a.i in this mark scheme extract

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

## Type MS · Find named items in a program

### Answer 8 of 18 · Question 3 part a.ii · 1 mark

2021 NOV · Paper 02 · extract 1 of 11 · find part a.ii in this mark scheme extract

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|
| 3 (a) (i) | A1 | C# and Python monthNumber/monthsAndDays/count/found (1)<br>Java monthNumber/monthsAndDays/count/found/input | | (1) |

## Type MS · Find named items in a program

### Answer 8 of 18 · Question 3 part a.ii · 1 mark

2021 NOV · Paper 02 · extract 2 of 11 · find part a.ii in this mark scheme extract

| | | | |
|---|---|---|---|
| 3 (a) (ii) | A2 | monthsAndDays (1) | (1) |

## Type MS · Find named items in a program

### Answer 8 of 18 · Question 3 part a.ii · 1 mark

2021 NOV · Paper 02 · extract 3 of 11 · find part a.ii in this mark scheme extract

| | | | |
|---|---|---|---|
| 3 (a) (iii) | A3 | C#<br>• WriteLine, ToString, ToInt32, ReadLine<br>Java<br>• Print, ParseInt, nextLine, ToString, Equals<br>Python<br>• print/len/int/input (1) | (1) |

## Type MS · Find named items in a program

### Answer 8 of 18 · Question 3 part a.ii · 1 mark

2021 NOV · Paper 02 · extract 4 of 11 · find part a.ii in this mark scheme extract

| | | | |
|---|---|---|---|
| 3 (a) (iv) | A4 | Display/showMonthNameAndDays (1) | (1) |

## Type MS · Find named items in a program

### Answer 8 of 18 · Question 3 part a.ii · 1 mark

2021 NOV · Paper 02 · extract 5 of 11 · find part a.ii in this mark scheme extract



## Type MS · Find named items in a program

### Answer 8 of 18 · Question 3 part a.ii · 1 mark

2021 NOV · Paper 02 · extract 6 of 11 · find part a.ii in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 3 (a) (vi) | A6<br>A7 | **Award 1 mark for each of:**<br><br>**Type of test data** / **Test data**<br>Boundary (1) / 1<br>Erroneous (1) / 15 | | (2) |

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

## Type MS · Find named items in a program

### Answer 8 of 18 · Question 3 part a.ii · 1 mark

2021 NOV · Paper 02 · extract 7 of 11 · find part a.ii in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 3 (b) | B1<br>B2 | **Award up to 2 marks for a linked explanation such as:**<br><br>• A library program would not know the names of the calling program's variables in advance (1) meaning it would not be a reusable solution (1)<br>• (Using parameters) enables reusable solutions (1) using different data values/arguments (1)<br>• Using parameters uses less memory / uses memory more efficiently (1) because it avoids the use of global variables / because the memory is freed after the subprogram is finished (1) | | (2) |

## Type MS · Find named items in a program

### Answer 8 of 18 · Question 3 part a.ii · 1 mark

2021 NOV · Paper 02 · extract 8 of 11 · find part a.ii in this mark scheme extract

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

## Type MS · Find named items in a program

### Answer 8 of 18 · Question 3 part a.ii · 1 mark

2021 NOV · Paper 02 · extract 9 of 11 · find part a.ii in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 3 (c) | **Award 1 mark for each of:** | | | |
| | C1 | Condition to check whether the year exactly divides by 4 or equivalent condition (1) | | |
| | C2 | Condition to check whether year exactly divides by 400 or equivalent condition (1) | | |
| | C3 | Condition to check whether year does not exactly divide by 100 or equivalent condition (1) | | |
| | C4 | Correct message for at least one of the conditions (1) | | |
| | C5 | Program executes and produces the correct output (1) | | |
| | C6 | Efficient solution (1) | | |
| | | | | (6) |
| Code examples | | | | |
| C# | | | | |
| Java | | | | |

```
if (year % 400 == 0)
 {
     Console.WriteLine(year + " is a leap year");
 }
else if (year % 4 == 0 && year % 100 !=0)
 {
     Console.WriteLine(year + " is a leap year");
 }
 else
 {
     Console.WriteLine(year + " is not a leap year");
 }
```

```
if (year % 400 == 0)
{
    System.out.print(year + " is a leap year");
}
else if (year % 4 == 0 && year % 100 !=0)
{
    System.out.print(year + " is a leap year");
}
else
{
    System.out.print(year + " is not a leap year");
}
```

## Type MS · Find named items in a program

### Answer 8 of 18 · Question 3 part a.ii · 1 mark

2021 NOV · Paper 02 · extract 10 of 11 · find part a.ii in this mark scheme extract

| | |
|---|---|
| Python | `if year % 400 == 0:`<br>`  print(year,"is a leap year")`<br>`elif year % 4 == 0 and year % 100 !=0:`<br>`  print(year,"is a lear year")`<br>`else:`<br>`  print(year,"is not a leap year")` |

## Type MS · Find named items in a program

### Answer 8 of 18 · Question 3 part a.ii · 1 mark

2021 NOV · Paper 02 · extract 11 of 11 · find part a.ii in this mark scheme extract

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

## Type MS · Find named items in a program

### Answer 9 of 18 · Question 3 part a.iii · 1 mark

2021 NOV · Paper 02 · extract 1 of 11 · find part a.iii in this mark scheme extract

| Question | Mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|
| 3 (a) (i) | A1 | C# and Python monthNumber/monthsAndDays/count/found (1)<br>Java monthNumber/monthsAndDays/count/found/input | | (1) |

## Type MS · Find named items in a program

### Answer 9 of 18 · Question 3 part a.iii · 1 mark

2021 NOV · Paper 02 · extract 2 of 11 · find part a.iii in this mark scheme extract

| | | | |
|---|---|---|---|
| 3 (a) (ii) | A2 | monthsAndDays (1) | (1) |

## Type MS · Find named items in a program

### Answer 9 of 18 · Question 3 part a.iii · 1 mark

2021 NOV · Paper 02 · extract 3 of 11 · find part a.iii in this mark scheme extract

| | | | |
|---|---|---|---|
| 3 (a) (iii) | A3 | C#<br>• WriteLine, ToString, ToInt32, ReadLine<br>Java<br>• Print, ParseInt, nextLine, ToString, Equals<br>Python<br>• print/len/int/input (1) | (1) |

## Type MS · Find named items in a program

### Answer 9 of 18 · Question 3 part a.iii · 1 mark

2021 NOV · Paper 02 · extract 4 of 11 · find part a.iii in this mark scheme extract

| | | | |
|---|---|---|---|
| 3 (a) (iv) | A4 | Display/showMonthNameAndDays (1) | (1) |

## Type MS · Find named items in a program

### Answer 9 of 18 · Question 3 part a.iii · 1 mark

2021 NOV · Paper 02 · extract 5 of 11 · find part a.iii in this mark scheme extract

| | | | |
|---|---|---|---|
| 3 (a) (v) | A5 | pMonth (1) | (1) |

## Type MS · Find named items in a program

### Answer 9 of 18 · Question 3 part a.iii · 1 mark

2021 NOV · Paper 02 · extract 6 of 11 · find part a.iii in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 3 (a) (vi) | A6<br>A7 | **Award 1 mark for each of:**<br><br>**Type of test data** / **Test data**<br>Boundary (1) / 1<br>Erroneous (1) / 15 | | (2) |

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

## Type MS · Find named items in a program

### Answer 9 of 18 · Question 3 part a.iii · 1 mark

2021 NOV · Paper 02 · extract 7 of 11 · find part a.iii in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 3 (b) | B1<br>B2 | **Award up to 2 marks for a linked explanation such as:**<br><br>• A library program would not know the names of the calling program's variables in advance (1) meaning it would not be a reusable solution (1)<br>• (Using parameters) enables reusable solutions (1) using different data values/arguments (1)<br>• Using parameters uses less memory / uses memory more efficiently (1) because it avoids the use of global variables / because the memory is freed after the subprogram is finished (1) | | (2) |

## Type MS · Find named items in a program

### Answer 9 of 18 · Question 3 part a.iii · 1 mark

2021 NOV · Paper 02 · extract 8 of 11 · find part a.iii in this mark scheme extract

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

## Type MS · Find named items in a program

### Answer 9 of 18 · Question 3 part a.iii · 1 mark

2021 NOV · Paper 02 · extract 9 of 11 · find part a.iii in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 3 (c) | **Award 1 mark for each of:** | | | |
| | C1 | Condition to check whether the year exactly divides by 4 or equivalent condition (1) | | |
| | C2 | Condition to check whether year exactly divides by 400 or equivalent condition (1) | | |
| | C3 | Condition to check whether year does not exactly divide by 100 or equivalent condition (1) | | |
| | C4 | Correct message for at least one of the conditions (1) | | |
| | C5 | Program executes and produces the correct output (1) | | |
| | C6 | Efficient solution (1) | | |
| | | | | (6) |
| Code examples | | | | |
| C# | | | | |
| Java | | | | |

```
if (year % 400 == 0)
 {
     Console.WriteLine(year + " is a leap year");
 }
else if (year % 4 == 0 && year % 100 !=0)
 {
     Console.WriteLine(year + " is a leap year");
 }
 else
 {
     Console.WriteLine(year + " is not a leap year");
 }
```

```
if (year % 400 == 0)
{
    System.out.print(year + " is a leap year");
}
else if (year % 4 == 0 && year % 100 !=0)
{
    System.out.print(year + " is a leap year");
}
else
{
    System.out.print(year + " is not a leap year");
}
```

## Type MS · Find named items in a program

### Answer 9 of 18 · Question 3 part a.iii · 1 mark

2021 NOV · Paper 02 · extract 10 of 11 · find part a.iii in this mark scheme extract

| | |
|---|---|
| Python | `if year % 400 == 0:`<br>`  print(year,"is a leap year")`<br>`elif year % 4 == 0 and year % 100 !=0:`<br>`  print(year,"is a lear year")`<br>`else:`<br>`  print(year,"is not a leap year")` |

## Type MS · Find named items in a program

### Answer 9 of 18 · Question 3 part a.iii · 1 mark

2021 NOV · Paper 02 · extract 11 of 11 · find part a.iii in this mark scheme extract

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

## Type MS · Find named items in a program

### Answer 10 of 18 · Question 3 part a.iv · 1 mark

2021 NOV · Paper 02 · extract 1 of 11 · find part a.iv in this mark scheme extract

| Question | | Answer | Additional Guidance | Mark |
|---|---|---|---|---|
| 3 (a) (i) | A1 | C# and Python monthNumber/monthsAndDays/count/found (1)<br>Java monthNumber/monthsAndDays/count/found/input | | (1) |

## Type MS · Find named items in a program

### Answer 10 of 18 · Question 3 part a.iv · 1 mark

2021 NOV · Paper 02 · extract 2 of 11 · find part a.iv in this mark scheme extract

| | | | |
|---|---|---|---|
| 3 (a) (ii) | A2 | monthsAndDays (1) | (1) |

## Type MS · Find named items in a program

### Answer 10 of 18 · Question 3 part a.iv · 1 mark

2021 NOV · Paper 02 · extract 3 of 11 · find part a.iv in this mark scheme extract

| | | | |
|---|---|---|---|
| 3 (a) (iii) | A3 | C#<br>• WriteLine, ToString, ToInt32, ReadLine<br>Java<br>• Print, ParseInt, nextLine, ToString, Equals<br>Python<br>• print/len/int/input (1) | (1) |

## Type MS · Find named items in a program

### Answer 10 of 18 · Question 3 part a.iv · 1 mark

2021 NOV · Paper 02 · extract 4 of 11 · find part a.iv in this mark scheme extract

| | | | |
|---|---|---|---|
| 3 (a) (iv) | A4 | Display/showMonthNameAndDays (1) | (1) |

## Type MS · Find named items in a program

### Answer 10 of 18 · Question 3 part a.iv · 1 mark

2021 NOV · Paper 02 · extract 5 of 11 · find part a.iv in this mark scheme extract

| | | | |
|---|---|---|---|
| 3 (a) (v) | A5 | pMonth (1) | (1) |

## Type MS · Find named items in a program

### Answer 10 of 18 · Question 3 part a.iv · 1 mark

2021 NOV · Paper 02 · extract 6 of 11 · find part a.iv in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 3 (a) (vi) | A6<br>A7 | **Award 1 mark for each of:**<br><br>**Type of test data** / **Test data**<br>Boundary (1) / 1<br>Erroneous (1) / 15 | | (2) |

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

## Type MS · Find named items in a program

### Answer 10 of 18 · Question 3 part a.iv · 1 mark

2021 NOV · Paper 02 · extract 7 of 11 · find part a.iv in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 3 (b) | B1<br>B2 | **Award up to 2 marks for a linked explanation such as:**<br><br>• A library program would not know the names of the calling program's variables in advance (1) meaning it would not be a reusable solution (1)<br>• (Using parameters) enables reusable solutions (1) using different data values/arguments (1)<br>• Using parameters uses less memory / uses memory more efficiently (1) because it avoids the use of global variables / because the memory is freed after the subprogram is finished (1) | | (2) |

## Type MS · Find named items in a program

### Answer 10 of 18 · Question 3 part a.iv · 1 mark

2021 NOV · Paper 02 · extract 8 of 11 · find part a.iv in this mark scheme extract

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

## Type MS · Find named items in a program

### Answer 10 of 18 · Question 3 part a.iv · 1 mark

2021 NOV · Paper 02 · extract 9 of 11 · find part a.iv in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 3 (c) | **Award 1 mark for each of:** | | | |
| | C1 | Condition to check whether the year exactly divides by 4 or equivalent condition (1) | | |
| | C2 | Condition to check whether year exactly divides by 400 or equivalent condition (1) | | |
| | C3 | Condition to check whether year does not exactly divide by 100 or equivalent condition (1) | | |
| | C4 | Correct message for at least one of the conditions (1) | | |
| | C5 | Program executes and produces the correct output (1) | | |
| | C6 | Efficient solution (1) | | |
| | | | | (6) |
| Code examples | | | | |
| C# | | | | |
| Java | | | | |

```
if (year % 400 == 0)
{
    Console.WriteLine(year + " is a leap year");
}
else if (year % 4 == 0 && year % 100 !=0)
{
    Console.WriteLine(year + " is a leap year");
}
else
{
    Console.WriteLine(year + " is not a leap year");
}
```

```
if (year % 400 == 0)
{
    System.out.print(year + " is a leap year");
}
else if (year % 4 == 0 && year % 100 !=0)
{
    System.out.print(year + " is a leap year");
}
else
{
    System.out.print(year + " is not a leap year");
}
```

## Type MS · Find named items in a program

### Answer 10 of 18 · Question 3 part a.iv · 1 mark

2021 NOV · Paper 02 · extract 10 of 11 · find part a.iv in this mark scheme extract

| | |
|---|---|
| Python | `if year % 400 == 0:`<br>`  print(year,"is a leap year")`<br>`elif year % 4 == 0 and year % 100 !=0:`<br>`  print(year,"is a lear year")`<br>`else:`<br>`  print(year,"is not a leap year")` |

## Type MS · Find named items in a program

### Answer 10 of 18 · Question 3 part a.iv · 1 mark

2021 NOV · Paper 02 · extract 11 of 11 · find part a.iv in this mark scheme extract

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

## Type MS · Find named items in a program

### Answer 11 of 18 · Question 3 part a.v · 1 mark

2021 NOV · Paper 02 · extract 1 of 11 · find part a.v in this mark scheme extract

| Question | Mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|
| 3 (a) (i) | A1 | C# and Python monthNumber/monthsAndDays/count/found (1)<br>Java monthNumber/monthsAndDays/count/found/input | | (1) |

## Type MS · Find named items in a program

### Answer 11 of 18 · Question 3 part a.v · 1 mark

2021 NOV · Paper 02 · extract 2 of 11 · find part a.v in this mark scheme extract

| | | | |
|---|---|---|---|
| 3 (a) (ii) | A2 | monthsAndDays (1) | (1) |

## Type MS · Find named items in a program

### Answer 11 of 18 · Question 3 part a.v · 1 mark

2021 NOV · Paper 02 · extract 3 of 11 · find part a.v in this mark scheme extract

| | | | |
|---|---|---|---|
| 3 (a) (iii) | A3 | C#<br>• WriteLine, ToString, ToInt32, ReadLine<br>Java<br>• Print, ParseInt, nextLine, ToString, Equals<br>Python<br>• print/len/int/input (1) | (1) |

## Type MS · Find named items in a program

### Answer 11 of 18 · Question 3 part a.v · 1 mark

2021 NOV · Paper 02 · extract 4 of 11 · find part a.v in this mark scheme extract

| | | | |
|---|---|---|---|
| 3 (a) (iv) | A4 | Display/showMonthNameAndDays (1) | (1) |

## Type MS · Find named items in a program

### Answer 11 of 18 · Question 3 part a.v · 1 mark

2021 NOV · Paper 02 · extract 5 of 11 · find part a.v in this mark scheme extract

| | | | |
|---|---|---|---|
| 3 (a) (v) | A5 | pMonth (1) | (1) |

## Type MS · Find named items in a program

### Answer 11 of 18 · Question 3 part a.v · 1 mark

2021 NOV · Paper 02 · extract 6 of 11 · find part a.v in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 3 (a) (vi) | A6<br>A7 | **Award 1 mark for each of:**<br><br>**Type of test data** / **Test data**<br>Boundary (1) / 1<br>Erroneous (1) / 15 | | (2) |

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

## Type MS · Find named items in a program

### Answer 11 of 18 · Question 3 part a.v · 1 mark

2021 NOV · Paper 02 · extract 7 of 11 · find part a.v in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 3 (b) | B1<br>B2 | **Award up to 2 marks for a linked explanation such as:**<br><br>• A library program would not know the names of the calling program's variables in advance (1) meaning it would not be a reusable solution (1)<br>• (Using parameters) enables reusable solutions (1) using different data values/arguments (1)<br>• Using parameters uses less memory / uses memory more efficiently (1) because it avoids the use of global variables / because the memory is freed after the subprogram is finished (1) | | (2) |

## Type MS · Find named items in a program

### Answer 11 of 18 · Question 3 part a.v · 1 mark

2021 NOV · Paper 02 · extract 8 of 11 · find part a.v in this mark scheme extract

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

## Type MS · Find named items in a program

### Answer 11 of 18 · Question 3 part a.v · 1 mark

2021 NOV · Paper 02 · extract 9 of 11 · find part a.v in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 3 (c) | **Award 1 mark for each of:** | | | |
| | C1 | Condition to check whether the year exactly divides by 4 or equivalent condition (1) | | |
| | C2 | Condition to check whether year exactly divides by 400 or equivalent condition (1) | | |
| | C3 | Condition to check whether year does not exactly divide by 100 or equivalent condition (1) | | |
| | C4 | Correct message for at least one of the conditions (1) | | |
| | C5 | Program executes and produces the correct output (1) | | |
| | C6 | Efficient solution (1) | | |
| | | | | (6) |
| Code examples | | | | |
| C# | ```<br>if (year % 400 == 0)<br>{<br>Console.WriteLine(year + " is a leap year");<br>}<br>else if (year % 4 == 0 && year % 100 !=0)<br>{<br>Console.WriteLine(year + " is a leap year");<br>}<br>else<br>{<br>Console.WriteLine(year + " is not a leap year");<br>}<br>// End of main program<br>``` | | | |
| Java | ```<br>if (year % 400 == 0)<br>{<br>System.out.print(year + " is a leap year");<br>}<br>else if (year % 4 == 0 && year % 100 !=0)<br>{<br>System.out.print(year + " is a leap year");<br>}<br>else<br>{<br>System.out.print(year + " is not a leap year");<br>}<br>``` | | | |

## Type MS · Find named items in a program

### Answer 11 of 18 · Question 3 part a.v · 1 mark

2021 NOV · Paper 02 · extract 10 of 11 · find part a.v in this mark scheme extract

| | |
|---|---|
| Python | `if year % 400 == 0:`<br>`  print(year,"is a leap year")`<br>`elif year % 4 == 0 and year % 100 !=0:`<br>`  print(year,"is a lear year")`<br>`else:`<br>`  print(year,"is not a leap year")` |

## Type MS · Find named items in a program

### Answer 11 of 18 · Question 3 part a.v · 1 mark

2021 NOV · Paper 02 · extract 11 of 11 · find part a.v in this mark scheme extract

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

## Type MS · Find named items in a program

### Answer 12 of 18 · Question 2 part a.iv · 1 mark

2022 JUN · Paper 02 · extract 1 of 7 · find part a.iv in this mark scheme extract



## Type MS · Find named items in a program

### Answer 12 of 18 · Question 2 part a.iv · 1 mark

2022 JUN · Paper 02 · extract 2 of 7 · find part a.iv in this mark scheme extract



## Type MS · Find named items in a program

### Answer 12 of 18 · Question 2 part a.iv · 1 mark

2022 JUN · Paper 02 · extract 3 of 7 · find part a.iv in this mark scheme extract



## Type MS · Find named items in a program

### Answer 12 of 18 · Question 2 part a.iv · 1 mark

2022 JUN · Paper 02 · extract 4 of 7 · find part a.iv in this mark scheme extract



## Type MS · Find named items in a program

### Answer 12 of 18 · Question 2 part a.iv · 1 mark

2022 JUN · Paper 02 · extract 5 of 7 · find part a.iv in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 2 (a)(v) | A5 | pNumber | | (1) |

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

## Type MS · Find named items in a program

### Answer 12 of 18 · Question 2 part a.iv · 1 mark

2022 JUN · Paper 02 · extract 6 of 7 · find part a.iv in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 2 (b) | **Award 1 mark for each of:** | | Logic of algorithm must be followed as set out.<br>Alternatives must address each point.<br>Do not penalise candidates who attempt more than the stated requirements.<br>Do not penalise spelling mistakes in the input message. | (11) |
| | **Evidence should be found in the Function** | | | |
| | B1 | Function interface is correct (1) | | |
| | B2 | Condition correct (for the number 1) (1) | | |
| | B3 | Loop initialised correctly (from 2 to the input number) (1) | | |
| | B4 | If condition correct (modulus) (1) | | |
| | B5 | Check returned (1) | | |
| | **Evidence should be found in the Main program** | | | |
| | B6 | Input of number stored as an integer variable (1) | | |
| | B7 | Input includes a space/colon / new line before the input value | | |
| | B8 | Function called correctly (1) | | |
| | B9 | Display includes number and message for either a prime number **OR** not (1) | | |
| | **Overall** | | | |
| | B10 | Compiling without syntax errors (1) | | |
| | B11 | Executing and producing the correct output (1) | | |
| Code examples | | | | |
| Python | | | | |

```python
# Write the function here
def checkPrime(pNumber):
    if pNumber == 1:
        check = False
    else:
        check = True
        for count in range(2,pNumber):
            if pNumber % count == 0 :
                check = False
    return check

# Write the main program here
number = int(input("Enter a number: "))
result = checkPrime(number)
if result == True:
    print(number,"is a prime number")
else:
    print(number,"is not a prime number")
```

## Type MS · Find named items in a program

### Answer 12 of 18 · Question 2 part a.iv · 1 mark

2022 JUN · Paper 02 · extract 7 of 7 · find part a.iv in this mark scheme extract

| Question | mp | | Additional Guidance | Mark |
|---|---|---|---|---|

## Type MS · Find named items in a program

### Answer 13 of 18 · Question 2 part a.v · 1 mark

2022 JUN · Paper 02 · extract 1 of 7 · find part a.v in this mark scheme extract



## Type MS · Find named items in a program

### Answer 13 of 18 · Question 2 part a.v · 1 mark

2022 JUN · Paper 02 · extract 2 of 7 · find part a.v in this mark scheme extract



## Type MS · Find named items in a program

### Answer 13 of 18 · Question 2 part a.v · 1 mark

2022 JUN · Paper 02 · extract 3 of 7 · find part a.v in this mark scheme extract



## Type MS · Find named items in a program

### Answer 13 of 18 · Question 2 part a.v · 1 mark

2022 JUN · Paper 02 · extract 4 of 7 · find part a.v in this mark scheme extract



## Type MS · Find named items in a program

### Answer 13 of 18 · Question 2 part a.v · 1 mark

2022 JUN · Paper 02 · extract 5 of 7 · find part a.v in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 2 (a)(v) | A5 | pNumber | | (1) |

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

## Type MS · Find named items in a program

### Answer 13 of 18 · Question 2 part a.v · 1 mark

2022 JUN · Paper 02 · extract 6 of 7 · find part a.v in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 2 (b) | **Award 1 mark for each of:** | | Logic of algorithm must be followed as set out.<br>Alternatives must address each point.<br>Do not penalise candidates who attempt more than the stated requirements.<br>Do not penalise spelling mistakes in the input message. | (11) |
| | **Evidence should be found in the Function** | | | |
| | B1 | Function interface is correct (1) | | |
| | B2 | Condition correct (for the number 1) (1) | | |
| | B3 | Loop initialised correctly (from 2 to the input number) (1) | | |
| | B4 | If condition correct (modulus) (1) | | |
| | B5 | Check returned (1) | | |
| | **Evidence should be found in the Main program** | | | |
| | B6 | Input of number stored as an integer variable (1) | | |
| | B7 | Input includes a space/colon / new line before the input value | | |
| | B8 | Function called correctly (1) | | |
| | B9 | Display includes number and message for either a prime number **OR** not (1) | | |
| | **Overall** | | | |
| | B10 | Compiling without syntax errors (1) | | |
| | B11 | Executing and producing the correct output (1) | | |
| Code examples | | | | |
| Python | | | | |

```python
# Write the function here
def checkPrime(pNumber):
    if pNumber == 1:
        check = False
    else:
        check = True
        for count in range(2,pNumber):
            if pNumber % count == 0 :
                check = False
    return check

# Write the main program here
number = int(input("Enter a number: "))
result = checkPrime(number)
if result == True:
    print(number,"is a prime number")
else:
    print(number,"is not a prime number")
```

## Type MS · Find named items in a program

### Answer 13 of 18 · Question 2 part a.v · 1 mark

2022 JUN · Paper 02 · extract 7 of 7 · find part a.v in this mark scheme extract

| Question | mp | | Additional Guidance | Mark |
|---|---|---|---|---|

## Type MS · Find named items in a program

### Answer 14 of 18 · Question 1 part b.iii · 1 mark

2023 JUN · Paper 02 · extract 1 of 3 · find part b.iii in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 1(a) | A1<br>A2<br>A3 | Award **one** mark for each correct cell:<br><br>Input: Debit / Credit / Description<br>Processing: Balance + Credit / Balance - Debit<br>Output: Balance | Input: accept examples of inputs e.g. 10.23 (must be in Figure 1)<br><br>Processing; mark first 3 things (ignore extra text) **if** correct e.g.<br>**balance + credit** – debit (1)<br>**balance + credit** + debit (1)<br>balance – credit (0)<br>accept examples from table e.g. 128.35 – 10.23 (1)<br><br>Output: accept description of formula e.g. subtract debit from balance (1) | **3** |

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|
| | | Award **one** mark for each of: | | |
| 1(b)(i) | B1 | This is 5% | | **1** |
| 1(b)(ii) | B2 | or<br>Accept \|\| for C# and Java | | **1** |
| 1(b)(iii) | B3 | grossAmt<br>netAmt | | **1** |
| 1(b)(iv) | B4 | TAX | | **1** |
| 1(b)(v) | B5 | inGross | | **1** |

## Type MS · Find named items in a program

### Answer 14 of 18 · Question 1 part b.iii · 1 mark

2023 JUN · Paper 02 · extract 2 of 3 · find part b.iii in this mark scheme extract

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|
| 1(c)(i) | C1 | Award **one** mark for:<br><br>To store a value that will not change (during program execution) (1) | | **1** |
| 1(c)(ii) | C2 | Award **one** mark for any of the following:<br><br>• It is less likely to be changed by accident (translation will error if attempt assignment)(1)<br>• The value only needs to be changed in one place (for maintenance) (1)<br>• It is easier to read/understand **the logic** of (1)<br>• Ensure consistency of value if it has a name rather than the hard-coded same value repeatedly (typo) (1)<br>• Can make code more efficient (optimisation / value stored rather than reference) | MP1 And MP4 are referring to programmer. Do **not** accept answers relating to user. | **1** |

## Type MS · Find named items in a program

### Answer 14 of 18 · Question 1 part b.iii · 1 mark

2023 JUN · Paper 02 · extract 3 of 3 · find part b.iii in this mark scheme extract

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

## Type MS · Find named items in a program

### Answer 15 of 18 · Question 1 part b.iv · 1 mark

2023 JUN · Paper 02 · extract 1 of 3 · find part b.iv in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 1(a) | A1<br>A2<br>A3 | Award **one** mark for each correct cell:<br><br>Input: Debit / Credit / Description<br>Processing: Balance + Credit / Balance - Debit<br>Output: Balance | Input: accept examples of inputs e.g. 10.23 (must be in Figure 1)<br><br>Processing; mark first 3 things (ignore extra text) **if** correct e.g.<br>**balance + credit** – debit (1)<br>**balance + credit** + debit (1)<br>balance – credit (0)<br>accept examples from table e.g. 128.35 – 10.23 (1)<br><br>Output: accept description of formula e.g. subtract debit from balance (1) | **3** |

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|
| | | Award **one** mark for each of: | | |
| 1(b)(i) | B1 | This is 5% | | **1** |
| 1(b)(ii) | B2 | or<br>Accept \|\| for C# and Java | | **1** |
| 1(b)(iii) | B3 | grossAmt<br>netAmt | | **1** |
| 1(b)(iv) | B4 | TAX | | **1** |
| 1(b)(v) | B5 | inGross | | **1** |

## Type MS · Find named items in a program

### Answer 15 of 18 · Question 1 part b.iv · 1 mark

2023 JUN · Paper 02 · extract 2 of 3 · find part b.iv in this mark scheme extract

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|
| 1(c)(i) | C1 | Award **one** mark for:<br><br>To store a value that will not change (during program execution) (1) | | **1** |
| 1(c)(ii) | C2 | Award **one** mark for any of the following:<br><br>• It is less likely to be changed by accident (translation will error if attempt assignment)(1)<br>• The value only needs to be changed in one place (for maintenance) (1)<br>• It is easier to read/understand **the logic** of (1)<br>• Ensure consistency of value if it has a name rather than the hard-coded same value repeatedly (typo) (1)<br>• Can make code more efficient (optimisation / value stored rather than reference) | MP1 And MP4 are referring to programmer. Do **not** accept answers relating to user. | **1** |

## Type MS · Find named items in a program

### Answer 15 of 18 · Question 1 part b.iv · 1 mark

2023 JUN · Paper 02 · extract 3 of 3 · find part b.iv in this mark scheme extract

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

## Type MS · Find named items in a program

### Answer 16 of 18 · Question 1 part b.v · 1 mark

2023 JUN · Paper 02 · extract 1 of 3 · find part b.v in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 1(a) | A1<br>A2<br>A3 | Award **one** mark for each correct cell:<br><br>Input: Debit / Credit / Description<br>Processing: Balance + Credit / Balance - Debit<br>Output: Balance | Input: accept examples of inputs e.g. 10.23 (must be in Figure 1)<br><br>Processing; mark first 3 things (ignore extra text) **if** correct e.g.<br>**balance + credit** – debit (1)<br>**balance + credit** + debit (1)<br>balance – credit (0)<br>accept examples from table e.g. 128.35 – 10.23 (1)<br><br>Output: accept description of formula e.g. subtract debit from balance (1) | **3** |

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|
| | | Award **one** mark for each of: | | |
| 1(b)(i) | B1 | This is 5% | | **1** |
| 1(b)(ii) | B2 | or<br>Accept \|\| for C# and Java | | **1** |
| 1(b)(iii) | B3 | grossAmt<br>netAmt | | **1** |
| 1(b)(iv) | B4 | TAX | | **1** |
| 1(b)(v) | B5 | inGross | | **1** |

## Type MS · Find named items in a program

### Answer 16 of 18 · Question 1 part b.v · 1 mark

2023 JUN · Paper 02 · extract 2 of 3 · find part b.v in this mark scheme extract

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|
| 1(c)(i) | C1 | Award **one** mark for:<br><br>To store a value that will not change (during program execution) (1) | | **1** |
| 1(c)(ii) | C2 | Award **one** mark for any of the following:<br><br>• It is less likely to be changed by accident (translation will error if attempt assignment)(1)<br>• The value only needs to be changed in one place (for maintenance) (1)<br>• It is easier to read/understand **the logic** of (1)<br>• Ensure consistency of value if it has a name rather than the hard-coded same value repeatedly (typo) (1)<br>• Can make code more efficient (optimisation / value stored rather than reference) | MP1 And MP4 are referring to programmer. Do **not** accept answers relating to user. | **1** |

## Type MS · Find named items in a program

### Answer 16 of 18 · Question 1 part b.v · 1 mark

2023 JUN · Paper 02 · extract 3 of 3 · find part b.v in this mark scheme extract

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

## Type MS · Find named items in a program

### Answer 17 of 18 · Question 1 part d.i · 1 mark

2025 JUN · Paper 02 · extract 1 of 5 · find part d.i in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 1(a) | A1 | **The only correct answer is B**<br><br>***A*** *is not correct because for is used in iteration*<br>***C*** *is not correct because new is used to declare an object*<br>***D*** *is not correct because while is used in repetition* | | **1** |

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|
| 1(b)(i) | B1<br>B2<br>B3 | Award one mark for each correct cell: | | **3** |

| Description | Syntax error | Logic error | Runtime error |
|---|---|---|---|
| The program does not produce the expected output. | | ✔ | |
| The program does not translate. | ✔ | | |
| The program crashes during execution. | | | ✔ |

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## Type MS · Find named items in a program

### Answer 17 of 18 · Question 1 part d.i · 1 mark

2025 JUN · Paper 02 · extract 2 of 5 · find part d.i in this mark scheme extract

Paper 2 – Application of Computational Thinking

| Question | mp | Answer | Additional guidance | Mark |
|---|---|---|---|---|
| 1(b)(ii) | B4 | Line 15 - : added to end of line | | |
| | B5 | Line 20 – 0 changed to 1 | | |
| | B6 | Line 22 – missing comma added between string and greenCount | | 3 |

```
1  # Q01bii
2
3  # initialise counters
4  ballCount = 0
5  greenCount = 0
6  redCount = 0
7
8  ballArray = ["green", "red", "green", "green", "red", "green","green", "green",
9              "green", "green", "green", "red", "green", "red", "green", "red",
10             "red", "green", "green", "red", "red", "green", "green", "red",
11             "green", "red", "red", "green", "green", "red", "green", "green",
12             "red", "green", "green", "green", "red", "green", "red","green"]
13
14 # iterate over ball array
15 for ball in ballArray:
16     ballCount = ballCount + 1
17     if ball == "green":
18         greenCount = greenCount +1
19     else:
20         redCount = redCount + 1
21
22 print(" Green: ", greenCount, "balls \n", "Red: ", redCount,
23       "balls \n", "Total: ", ballCount, "balls")
```

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# Type MS · Find named items in a program

## Answer 17 of 18 · Question 1 part d.i · 1 mark

2025 JUN · Paper 02 · extract 3 of 5 · find part d.i in this mark scheme extract

Paper 2 – Application of Computational Thinking

| Question | mp | Answer | Additional guidance | Mark |
|---|---|---|---|---|
| 1(c)(i) | C1 | Award one mark for:<br>• To store a value that must not be changed/prevent accidental change of value<br>• To simplify code maintenance/so that if the value must be updated the change only has to made in a single place<br>• To avoid inconsistency error of specifying same value multiple times<br>• To make code more readable/give meaningful names to values | Accidental change cannot refer to 'user' | 1 |

| Question | mp | Answer | Additional guidance | Mark |
|---|---|---|---|---|
| 1(c)(ii) | C2<br>C3 | Award **two** marks for a linked description such as:<br>• A variable stores a single value (1) whereas an array stores multiple values (1) | | 2 |

| Question | mp | Answer | Additional guidance | Mark |
|---|---|---|---|---|
| 1(d)(i) | D1 | Award **one** mark for:<br>• getWages (1) | Accept 'main'<br>Accept:<br>def getWages<br>def getWages (inSales, inAge)<br>line 7 (Python) | 1 |

3 of 23

## Type MS · Find named items in a program

### Answer 17 of 18 · Question 1 part d.i · 1 mark

2025 JUN · Paper 02 · extract 4 of 5 · find part d.i in this mark scheme extract

Paper 2 – Application of Computational Thinking

| Question | mp | Answer | Additional guidance | Mark |
|---|---|---|---|---|
| 1(d)(ii) | D2 | Award **one** mark for:<br>• and (1) | Do not accept line numbers<br>Accept:<br>&& | 1 |

| Question | mp | Answer | Additional guidance | Mark |
|---|---|---|---|---|
| 1(d)(iii) | D3 | Award **one** mark for any of the following:<br>• sales (1)<br>• wages (1)<br>• inSales (1) | Do not accept line numbers | 1 |

| Question | mp | Answer | Additional guidance | Mark |
|---|---|---|---|---|
| 1(d)(iv) | D4 | Award **one** mark for any of the following:<br>• inSales (1)<br>• inAge (1) | Do not accept line numbers | 1 |

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## Type MS · Find named items in a program

### Answer 17 of 18 · Question 1 part d.i · 1 mark

2025 JUN · Paper 02 · extract 5 of 5 · find part d.i in this mark scheme extract

Paper 2 – Application of Computational Thinking

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

## Type MS · Find named items in a program

### Answer 18 of 18 · Question 1 part d.iii · 1 mark

2025 JUN · Paper 02 · extract 1 of 5 · find part d.iii in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 1(a) | A1 | **The only correct answer is B**<br><br>***A*** *is not correct because for is used in iteration*<br>***C*** *is not correct because new is used to declare an object*<br>***D*** *is not correct because while is used in repetition* | | **1** |

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|
| 1(b)(i) | B1<br>B2<br>B3 | Award one mark for each correct cell: | | **3** |

| Description | Syntax error | Logic error | Runtime error |
|---|---|---|---|
| The program does not produce the expected output. | | ✔ | |
| The program does not translate. | ✔ | | |
| The program crashes during execution. | | | ✔ |

1 of 23

## Type MS · Find named items in a program

### Answer 18 of 18 · Question 1 part d.iii · 1 mark

2025 JUN · Paper 02 · extract 2 of 5 · find part d.iii in this mark scheme extract

Paper 2 – Application of Computational Thinking

| Question | mp | Answer | Additional guidance | Mark |
|---|---|---|---|---|
| 1(b)(ii) | B4 | Line 15 - : added to end of line | | |
| | B5 | Line 20 – 0 changed to 1 | | |
| | B6 | Line 22 – missing comma added between string and greenCount | | 3 |

```
# Q01bii

# initialise counters
ballCount = 0
greenCount = 0
redCount = 0

ballArray = ["green", "red", "green", "green", "red", "green","green", "green",
             "green", "green", "green", "red", "green", "red", "green", "red",
             "red", "green", "green", "red", "red", "green", "green", "red",
             "green", "red", "red", "green", "green", "red", "green", "green",
             "red", "green", "green", "green", "red", "green", "red","green"]

# iterate over ball array
for ball in ballArray:
    ballCount = ballCount + 1
    if ball == "green":
        greenCount = greenCount +1
    else:
        redCount = redCount + 1

print(" Green: ", greenCount, "balls \n", "Red: ", redCount,
      "balls \n", "Total: ", ballCount, "balls")
```

2 of 23

# Type MS · Find named items in a program

## Answer 18 of 18 · Question 1 part d.iii · 1 mark

2025 JUN · Paper 02 · extract 3 of 5 · find part d.iii in this mark scheme extract

Paper 2 – Application of Computational Thinking

| Question | mp | Answer | Additional guidance | Mark |
|---|---|---|---|---|
| 1(c)(i) | C1 | Award one mark for:<br>• To store a value that must not be changed/prevent accidental change of value<br>• To simplify code maintenance/so that if the value must be updated the change only has to made in a single place<br>• To avoid inconsistency error of specifying same value multiple times<br>• To make code more readable/give meaningful names to values | Accidental change cannot refer to 'user' | 1 |

| Question | mp | Answer | Additional guidance | Mark |
|---|---|---|---|---|
| 1(c)(ii) | C2<br>C3 | Award **two** marks for a linked description such as:<br>• A variable stores a single value (1) whereas an array stores multiple values (1) | | 2 |

| Question | mp | Answer | Additional guidance | Mark |
|---|---|---|---|---|
| 1(d)(i) | D1 | Award **one** mark for:<br>• getWages (1) | Accept 'main'<br>Accept:<br>def getWages<br>def getWages (inSales, inAge)<br>line 7 (Python) | 1 |

3 of 23

## Type MS · Find named items in a program

### Answer 18 of 18 · Question 1 part d.iii · 1 mark

2025 JUN · Paper 02 · extract 4 of 5 · find part d.iii in this mark scheme extract

Paper 2 – Application of Computational Thinking

| Question | mp | Answer | Additional guidance | Mark |
|---|---|---|---|---|
| 1(d)(ii) | D2 | Award **one** mark for:<br>• and (1) | Do not accept line numbers<br>Accept:<br>&& | 1 |

| Question | mp | Answer | Additional guidance | Mark |
|---|---|---|---|---|
| 1(d)(iii) | D3 | Award **one** mark for any of the following:<br>• sales (1)<br>• wages (1)<br>• inSales (1) | Do not accept line numbers | 1 |

| Question | mp | Answer | Additional guidance | Mark |
|---|---|---|---|---|
| 1(d)(iv) | D4 | Award **one** mark for any of the following:<br>• inSales (1)<br>• inAge (1) | Do not accept line numbers | 1 |

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## Type MS · Find named items in a program

### Answer 18 of 18 · Question 1 part d.iii · 1 mark

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Paper 2 – Application of Computational Thinking

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|